Example 3.55 Check the value of k for which the given system of equations $kx+2y=3;\ 2x-3y=1$ has a unique solution.

Solution Given linear equations are

$$kx+2y=3 \ .....(1) \qquad \left[\begin{matrix} a_1x+b_1y+c_1=0 \\ a_2x+b_2y+c_2=0 \end{matrix}\right]$$
$$2x-3y=1 \ .....(2)$$

Here $a_1=k,\ b_1=2,\ a_2=2,\ b_2=-3$;

For unique solution we take $\frac{a_1}{a_2} \uparrow \frac{b_1}{b_2}$; therefore $\frac{k}{2} \neq \frac{2}{-3}$; $k \neq \frac{4}{-3}$, that is $k \neq -\frac{4}{3}$.

Example 3.56 Find the value of k, for the following system of equation has infinitely many solutions. $2x-3y=7;\ (k+2)x-(2k+1)y=3(2k-1)$

Solution Given two linear equations are

$$2x-3y=7 \qquad \left[\begin{matrix} a_1x+b_1y+c_1=0 \\ a_2x+b_2y+c_2=0 \end{matrix}\right]$$
$$(k+2)x-(2k+1)y=3(2k-1)$$

Here $a_1=2,\ b_1=-3,\ a_2=(k+2),\ b_2=-(2k+1),\ c_1=7,\ c_2=3(2k-1)$

For infinite number of solution we consider $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$

$$\frac{2}{k+2}=\frac{-3}{-(2k+1)}=\frac{7}{3(2k-1)}$$

$$\frac{2}{k+2}=\frac{-3}{-(2k+1)}$$
$$2(2k+1)=3(k+2)$$
$$4k+2=3k+6$$
$$k=4$$

$$\frac{-3}{-(2k+1)}=\frac{7}{3(2k-1)}$$
$$9(2k-1)=7(2k+1)$$
$$18k-9=14k+7$$
$$4k=16$$
$$k=4$$

Example 3.57 Find the value of k for which the system of linear equations $8x+5y=9$; $kx+10y=15$ has no solution.

Solution Given linear equations are

$$8x+5y=9 \qquad \left[\begin{matrix} a_1x+b_1y+c_1=0 \\ a_2x+b_2y+c_2=0 \end{matrix}\right]$$
$$kx+10y=15$$

Here $a_1=8,\ b_1=5,\ c_1=9,\ a_2=k, b_2=10, c_2=15$

For no solution, we know that $\frac{a_1}{a_2}=\frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ and so, $\frac{8}{k}=\frac{5}{10} \neq \frac{9}{15}$

$$80=5k$$
$$k=16$$

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